



Wheel/Rail Damage Mechanisms

Richard Stock - Global Head of Rail Solutions

Plasser American / Plasser & Theurer

DALLAS, TX



Materials

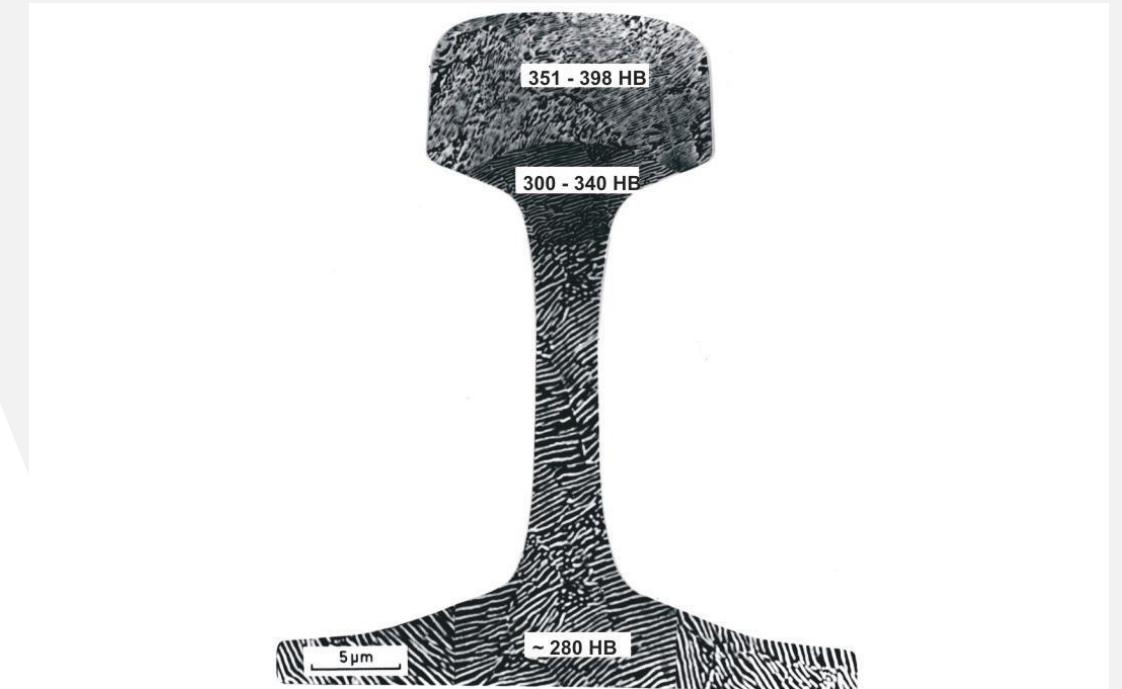
What Rails and Wheels Are
Made of





Steel: Rail and Wheel

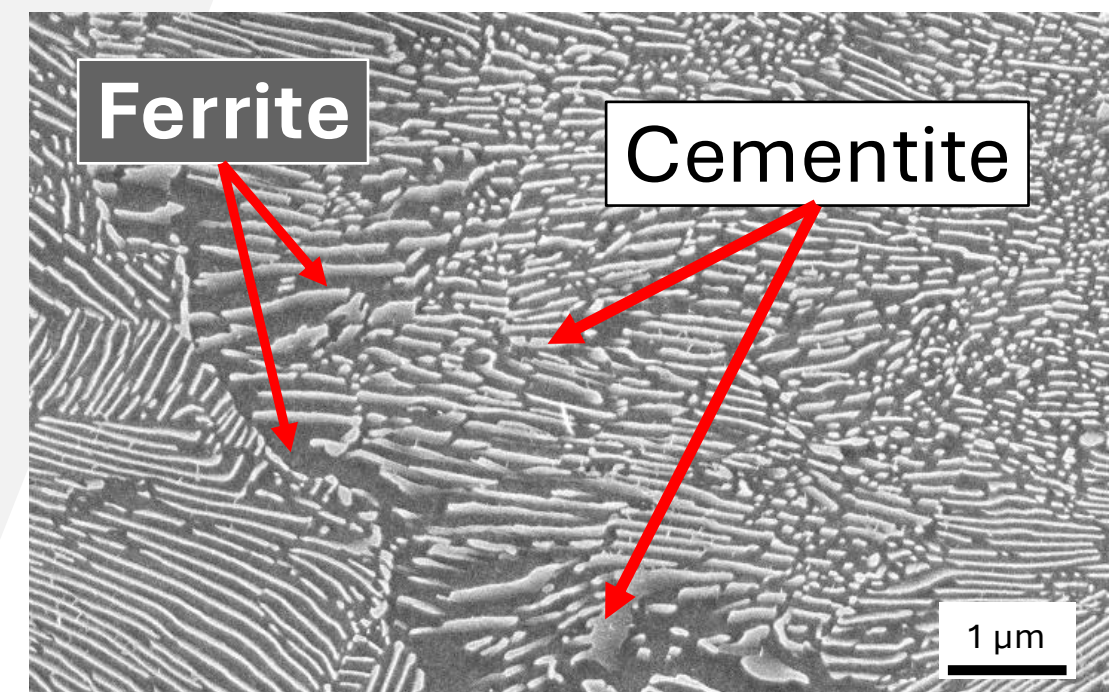
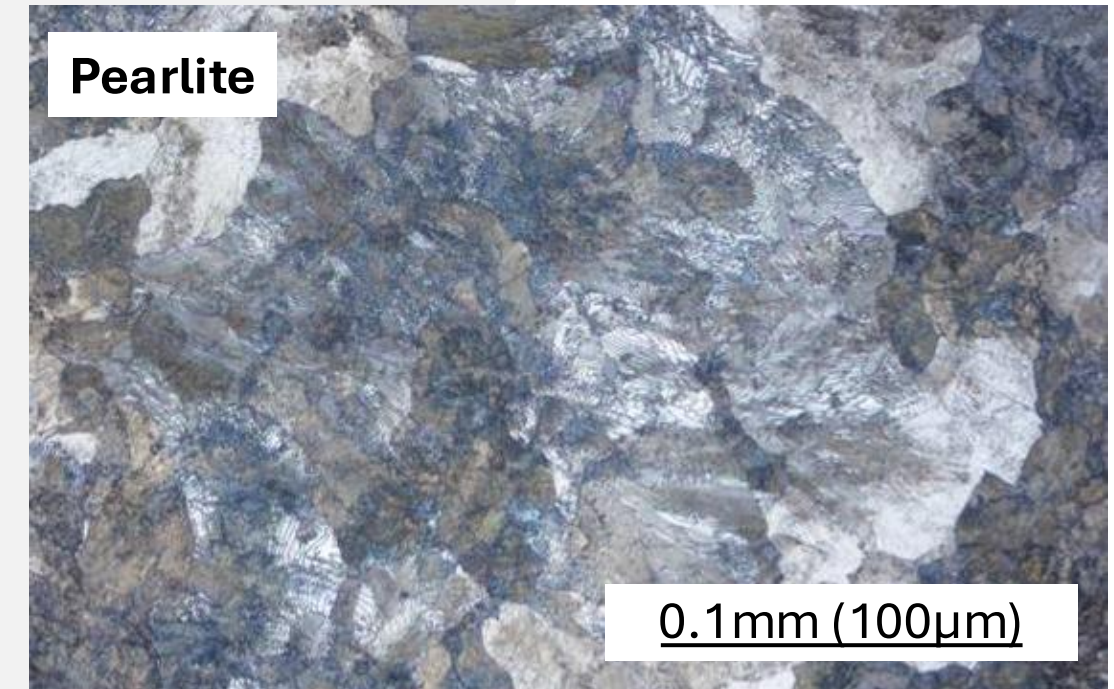
- Rail: Hardness: 200 – 450 BHN
- Wheel: Hardness: 200 – 415 BHN
- Sufficient Toughness and Strength
- Ductile material behaviour
- Sufficient electric conductivity
- Reasonable weldability
- Excellent machineability
- Reasonable price





Pearlitic Microstructure

- Two phase material:
 - Ferrite: very soft, $C_{\max} = 0.02\%$
 - Cementite: Fe_3C , very hard, $C = 6.67\%$
- Lamellar or layer structure
- Pure pearlitic structure at $0.77\% \text{ C}$ (Eutectoid point)
 - $C > 0.77\%$: Hypereutectoid Steel
- Lamella spacing defines hardness and strength without influencing the toughness





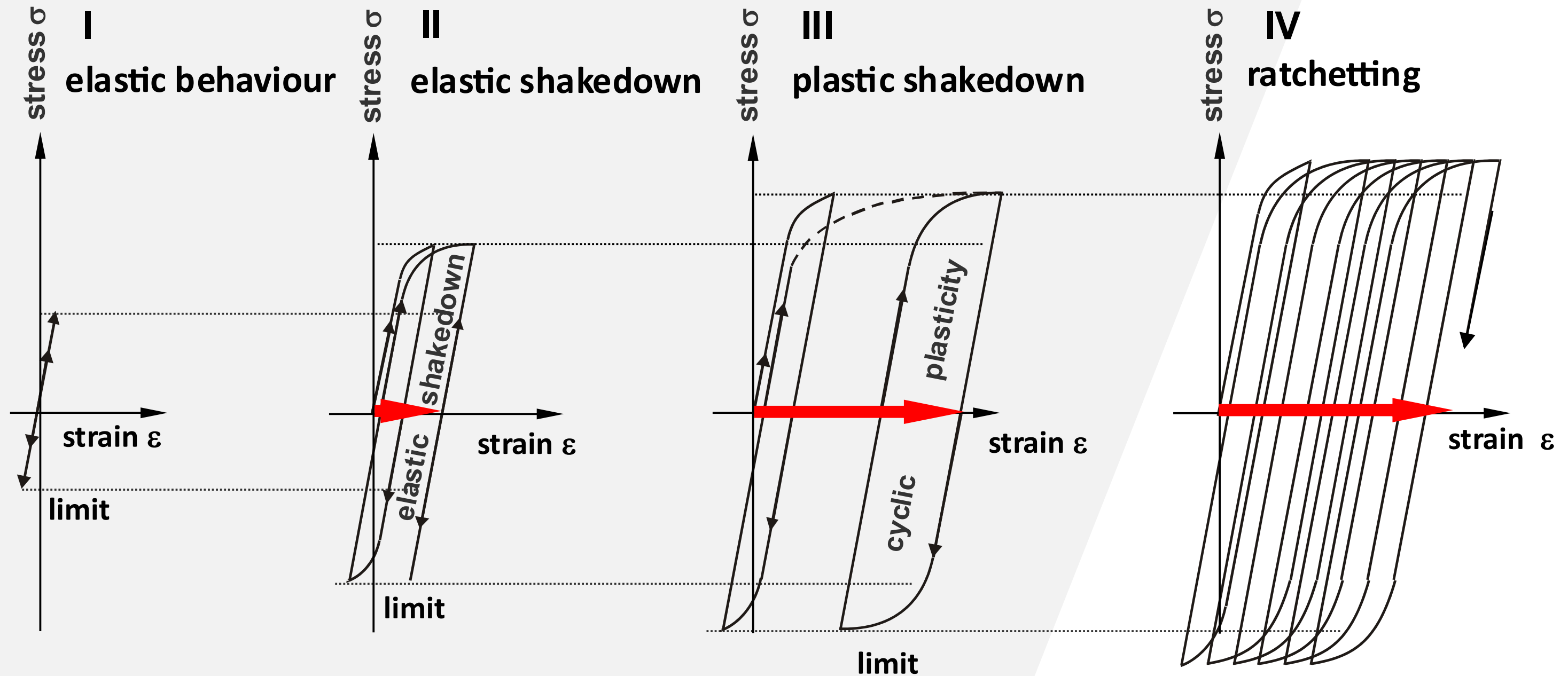
Wheel / Rail Damage Mechanisms

Material Reaction to Loading





Material Behaviour Under Load



→ accumulated plastic strain / plastic deformation



Plastic Deformation

- Contact loads always above elastic material limit.
 - On a microscopic scale close to the rail/wheel surface
 - Incremental accumulation of plastic deformation - ratchetting
- Plastic flow enclosed by bulk elastic material
 - No “global” plastic flow

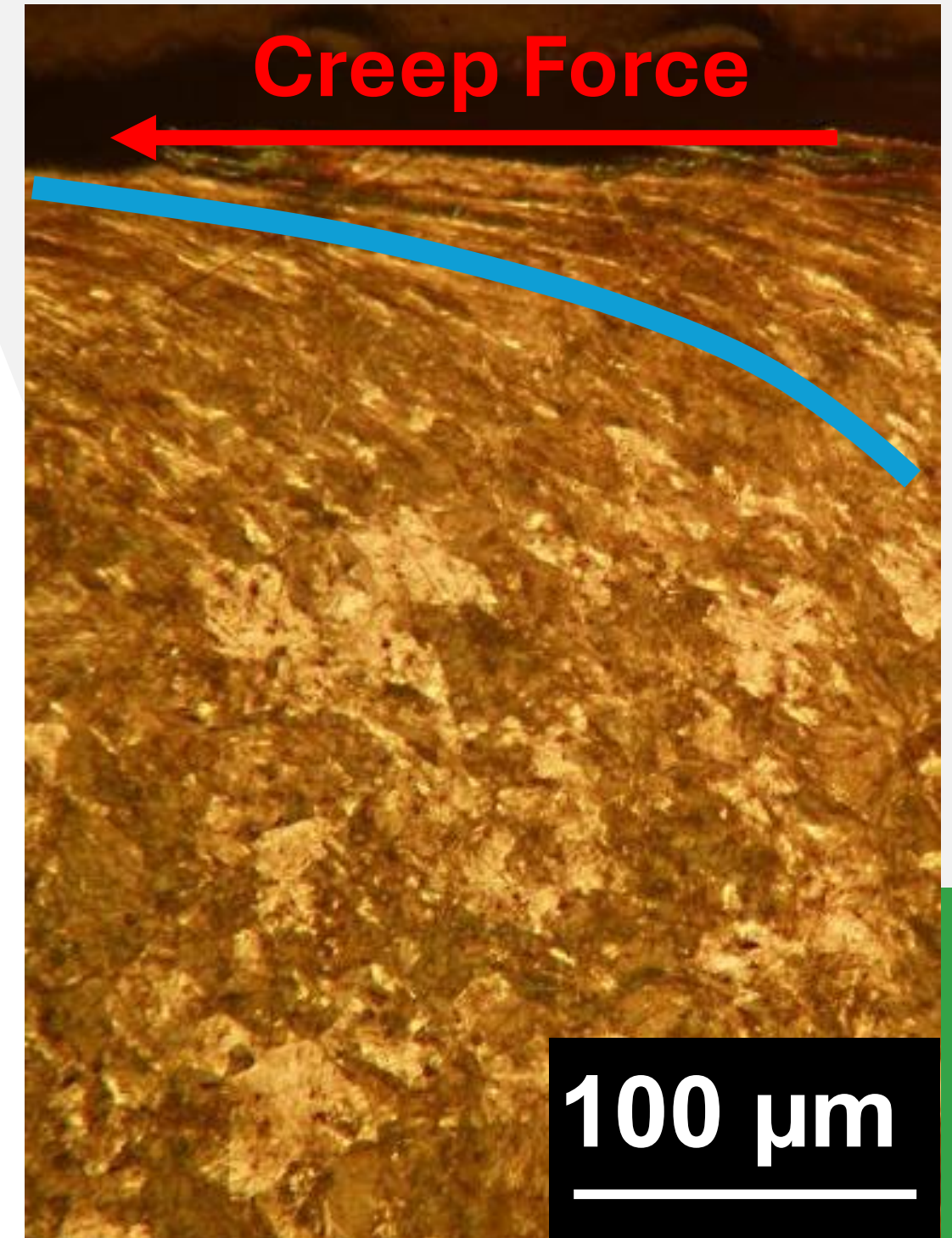
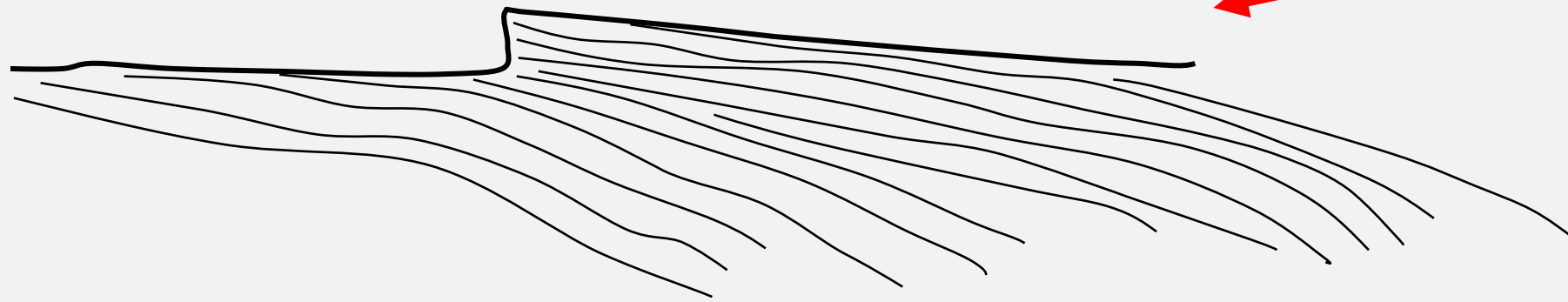


Photo: voestalpine



Material Response: Deformation

Severely deformed and aligned material structure at the rail/wheel surface



Non-deformed material structure

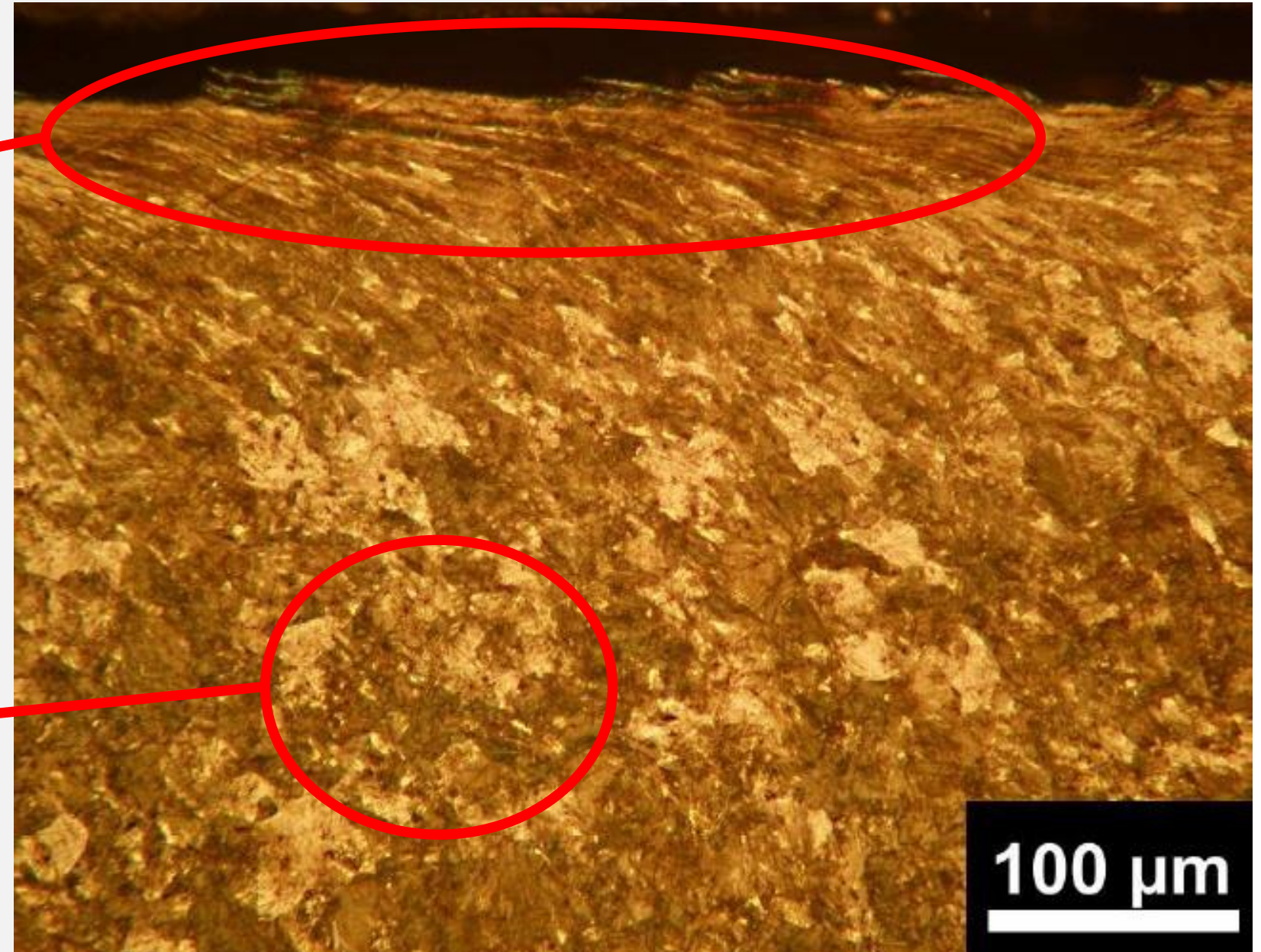
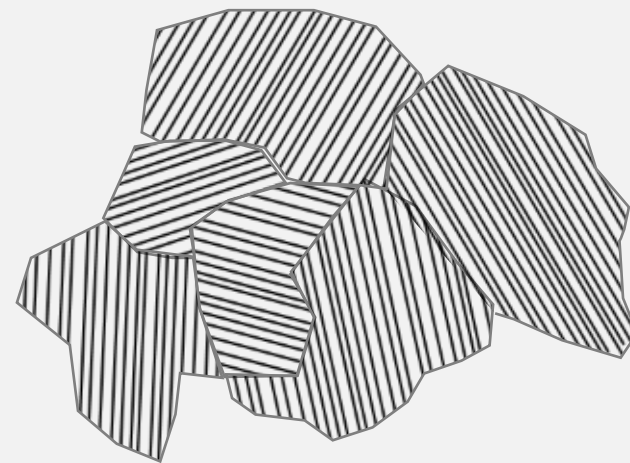


Photo: voestalpine



Plastic Deformation

- On a macroscopic scale – change of profile shape.
- Material flow – e.g. lipping

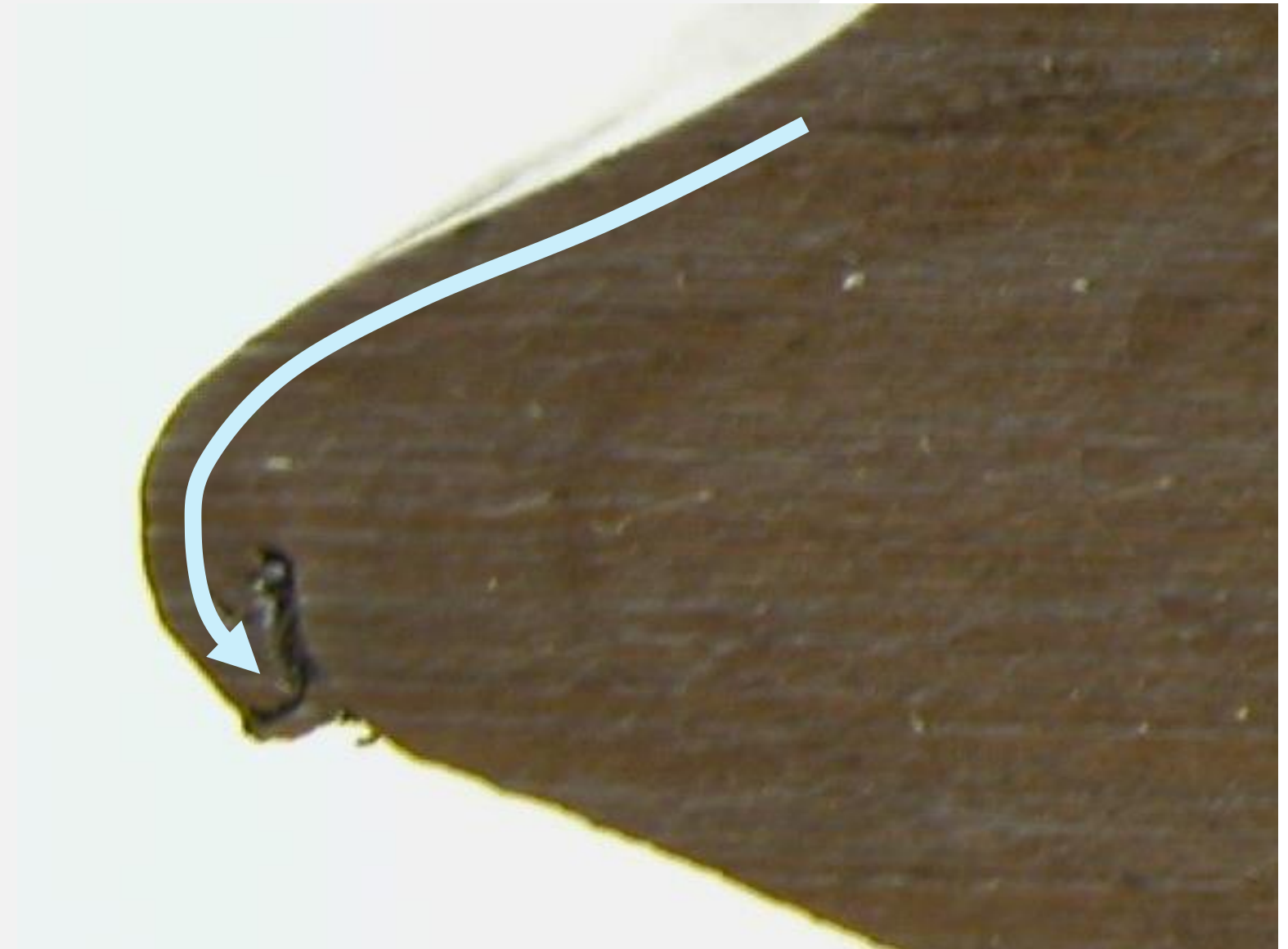


Photo: voestalpine



Material Wear

- Continuous material removal from rail/wheel surface due to interaction of wheel and rail.
- Several modes of wear
 - Adhesive wear
 - Abrasive wear
 - Fatigue wear
 - Corrosive wear
- Several types of wear
 - Natural Wear + Artificial Wear = Combined Wear



Photo by L.B. Foster

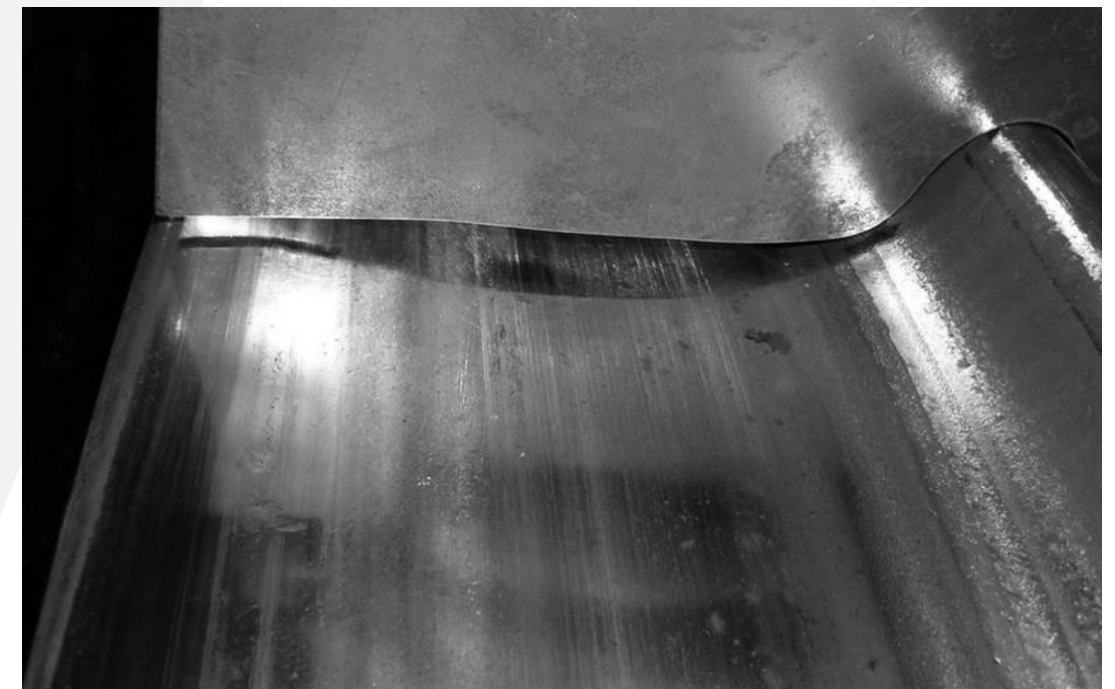
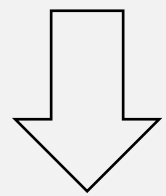
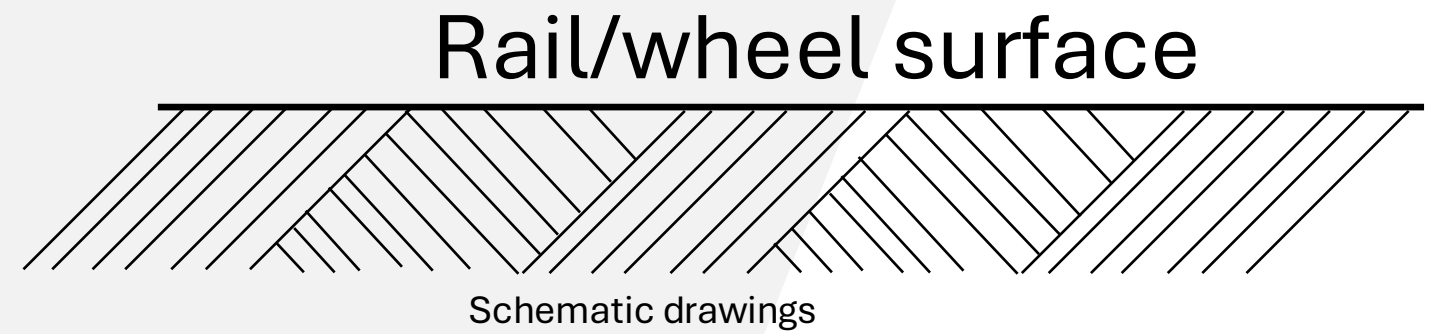


Photo by A. Eckberg



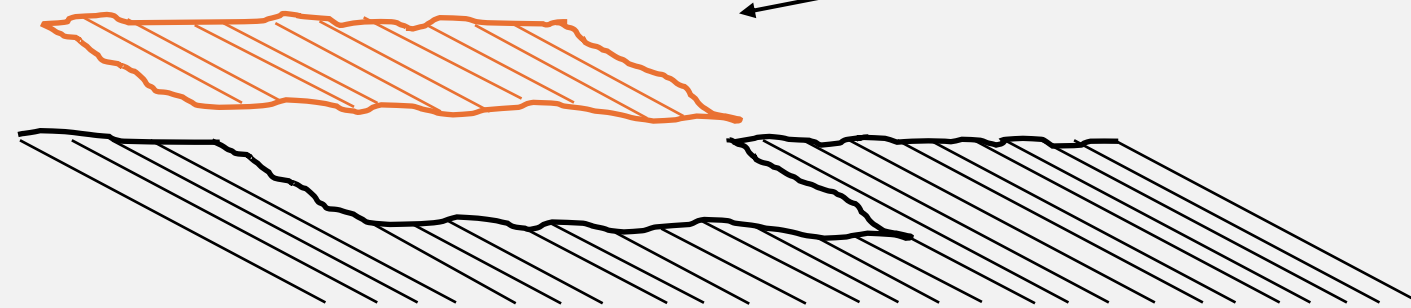
Material Response: Wear

Non-deformed, initial material condition

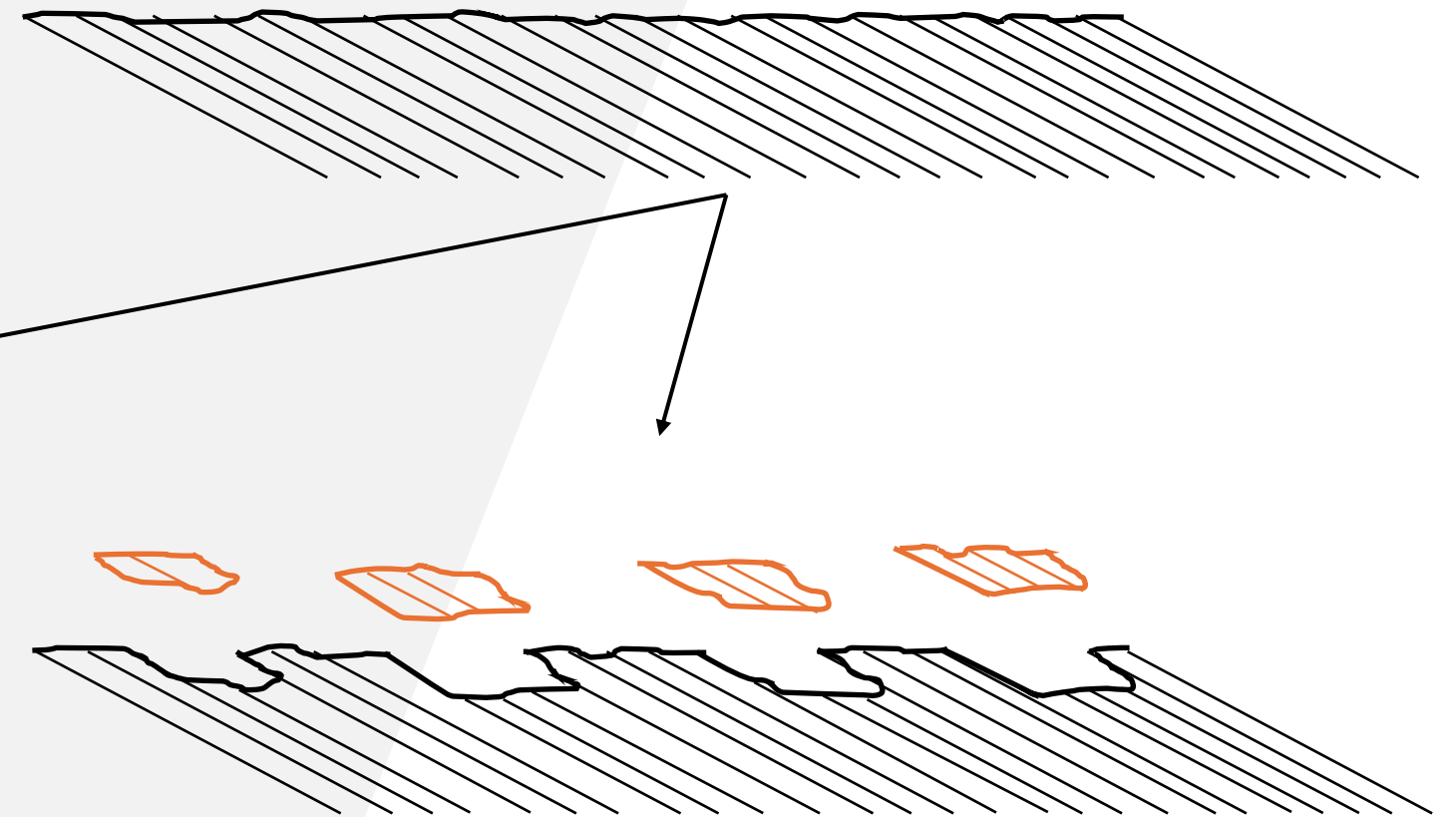


Loading conditions, material properties

Severely deformed rail/wheel surface



Severe wear

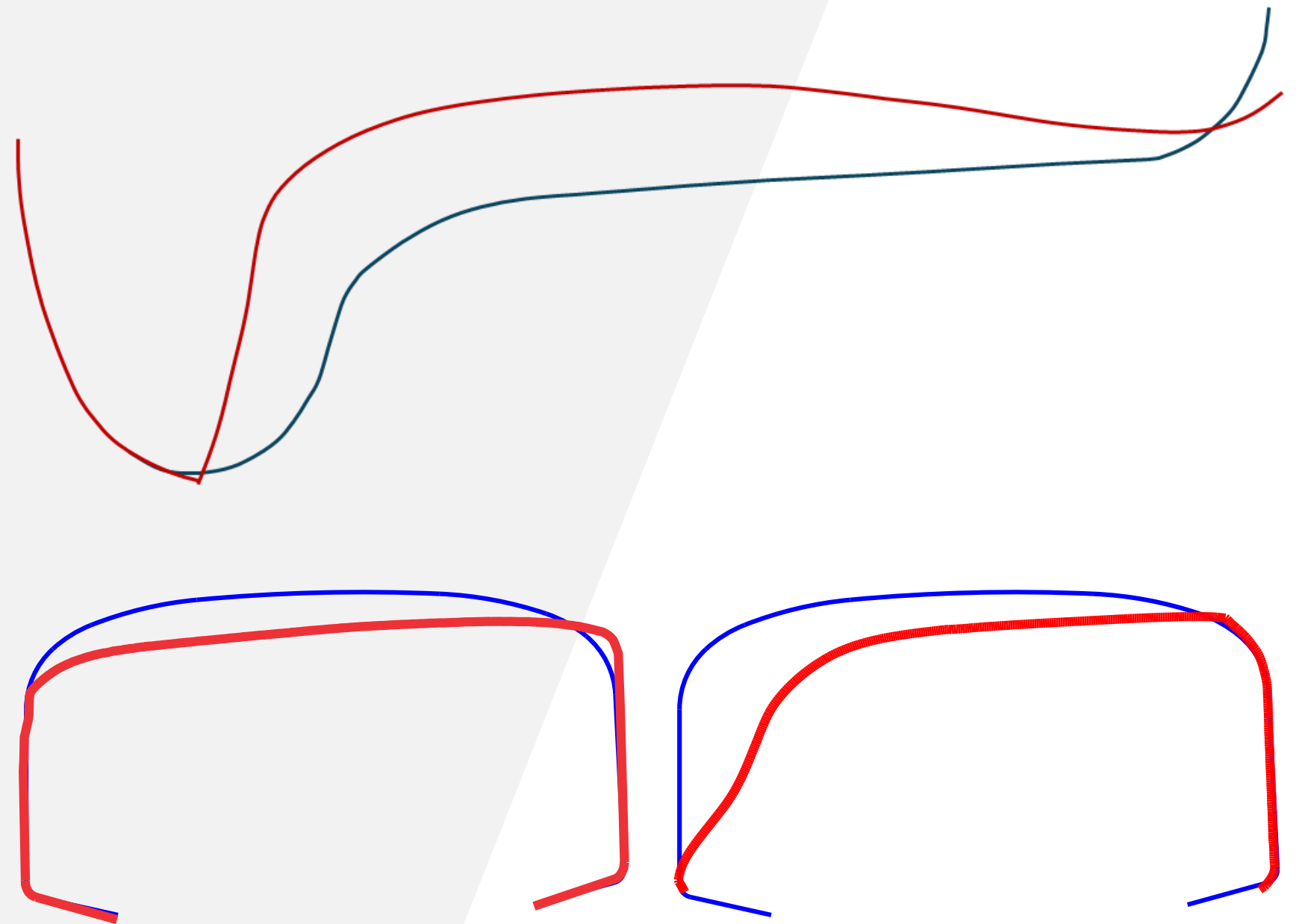


Mild wear



Wear effects: Profile Degradation

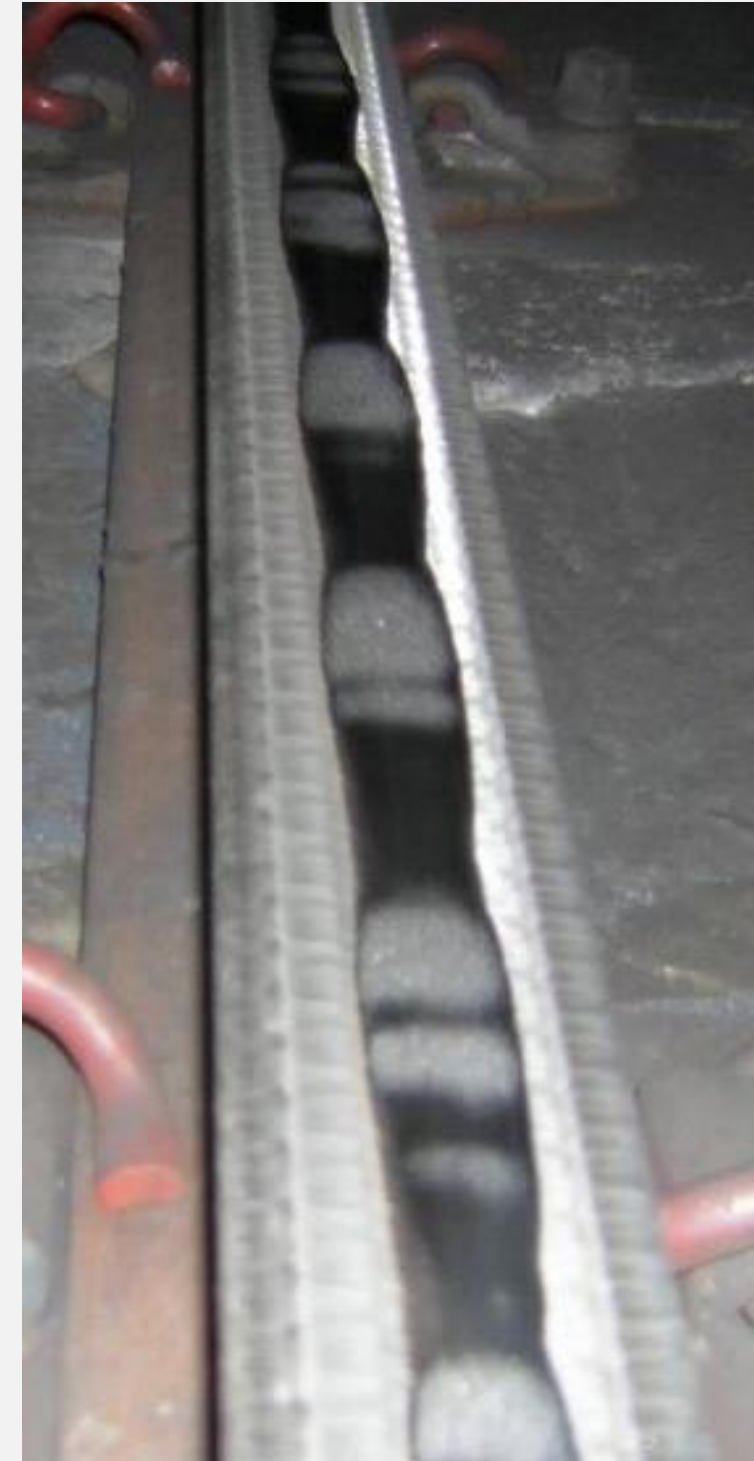
- Wheel:
 - Hollow Worn Wheel
 - Thin Flange
- Rail:
 - Flat Rail
 - Sever Gauge Wear





Rail: Corrugation

- Combination of wear and plastic flow
- Wave structure on the rail surface (tangent / curve)
- Short wave (25mm-80mm wavelength) or long wave (100-300mm) corrugation
- Multiple sub-classifications

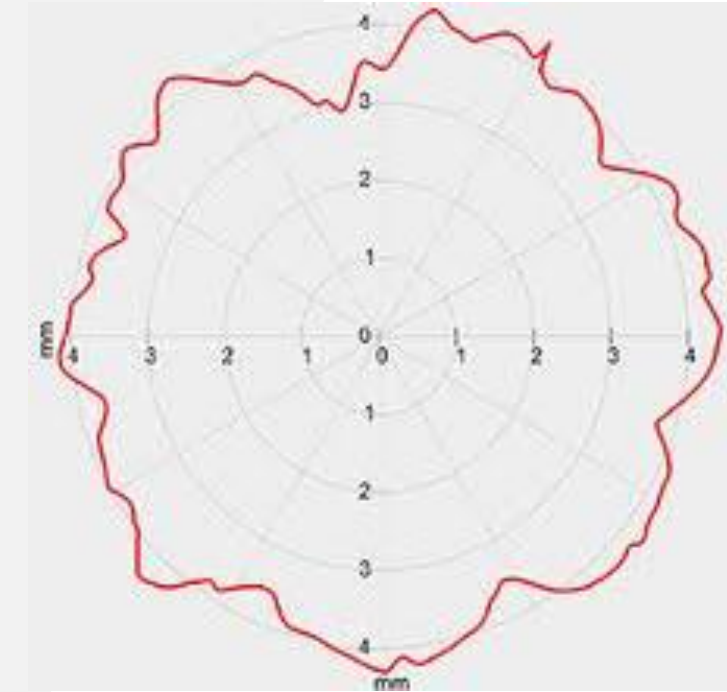


Photos by L.B. Foster



Out of Round Wheels

- Any deviation from a perfectly round wheel
- Polygonised wheels (periodic) caused by:
 - Stick-Slip Effects
 - Re-Profiling
 - Dynamic wheel-set oscillations caused by resonances in vehicle/track interaction.
- Wheel flat (discrete)
 - Wear during sliding of a locked wheel

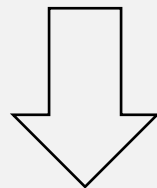
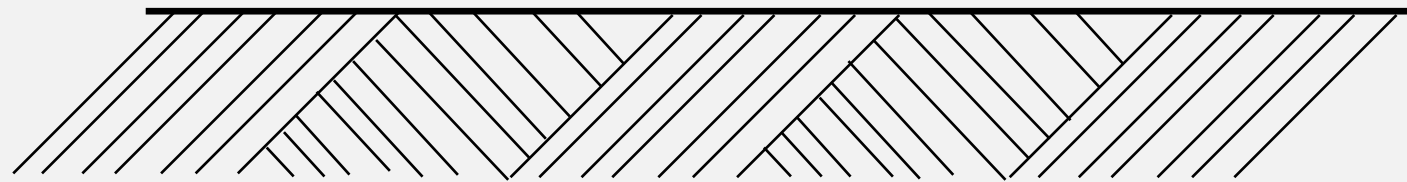


Source: Wikipedia, Bobo11,
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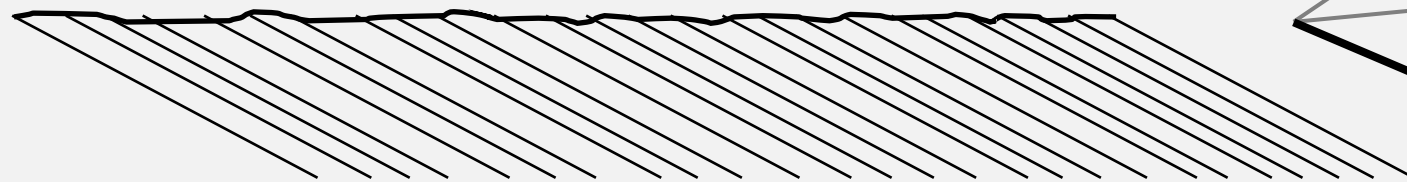


Material Response: Cracks

Non-deformed, initial material condition

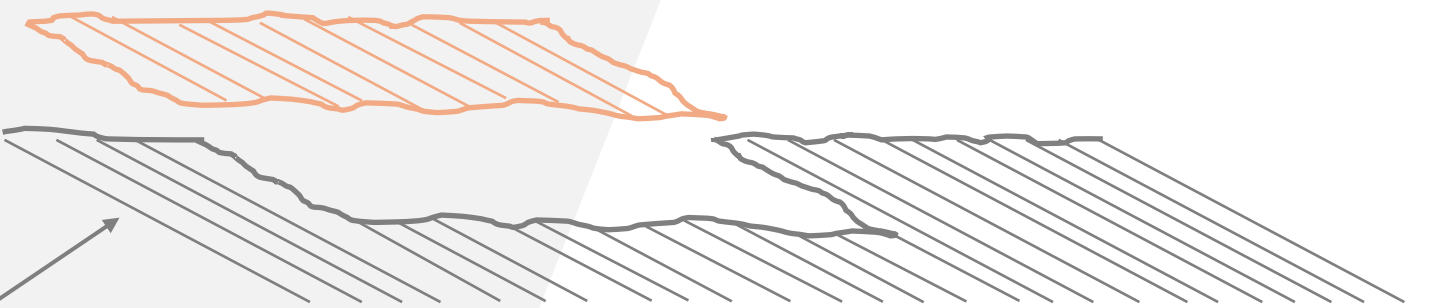


Severely deformed surface

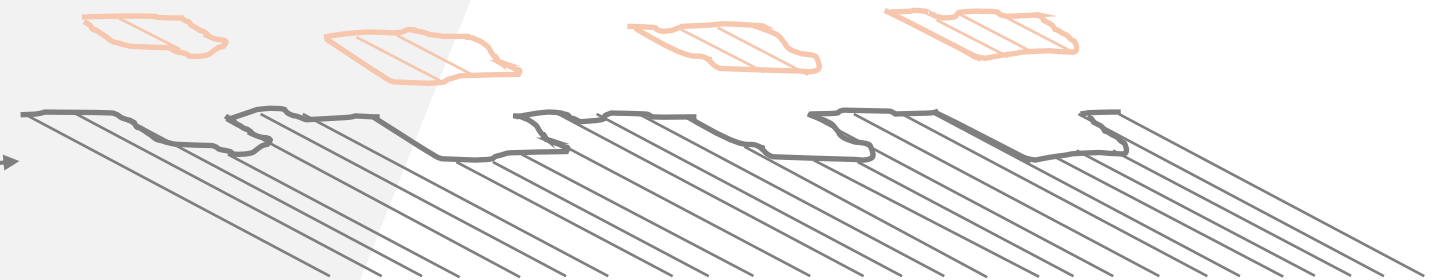


Schematic drawings

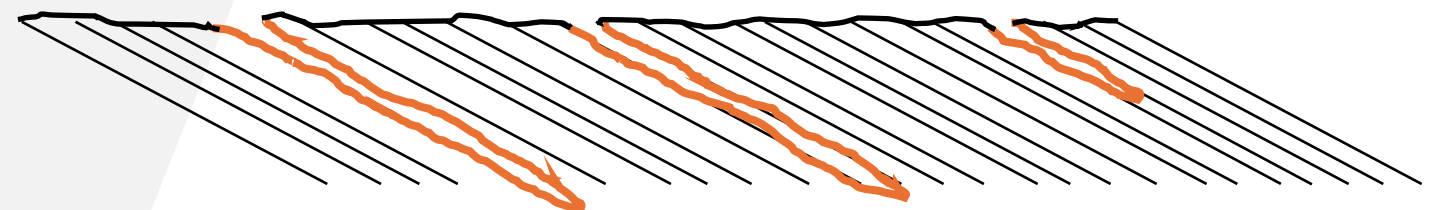
Severe wear



Mild wear

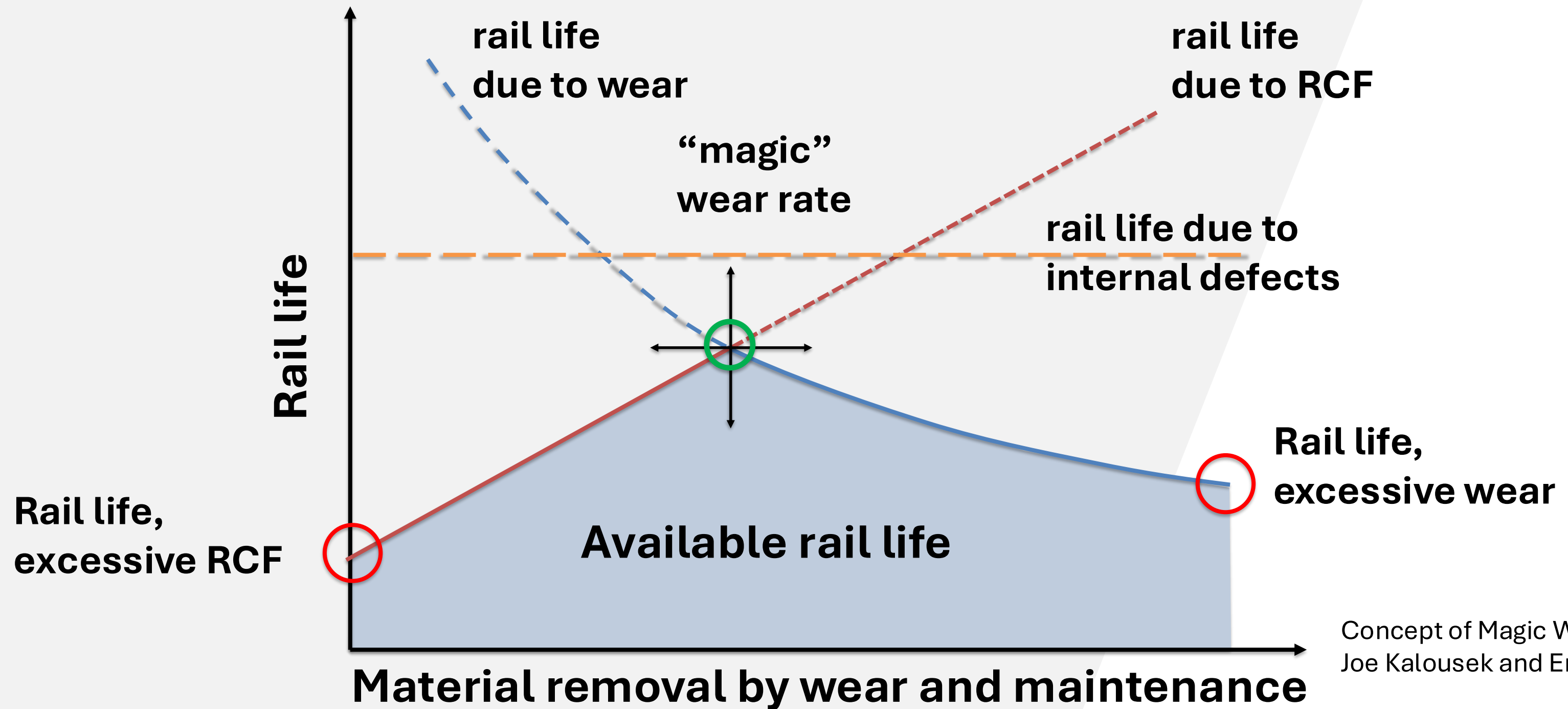


Surface cracks





Magic Wear Rate (Rail)

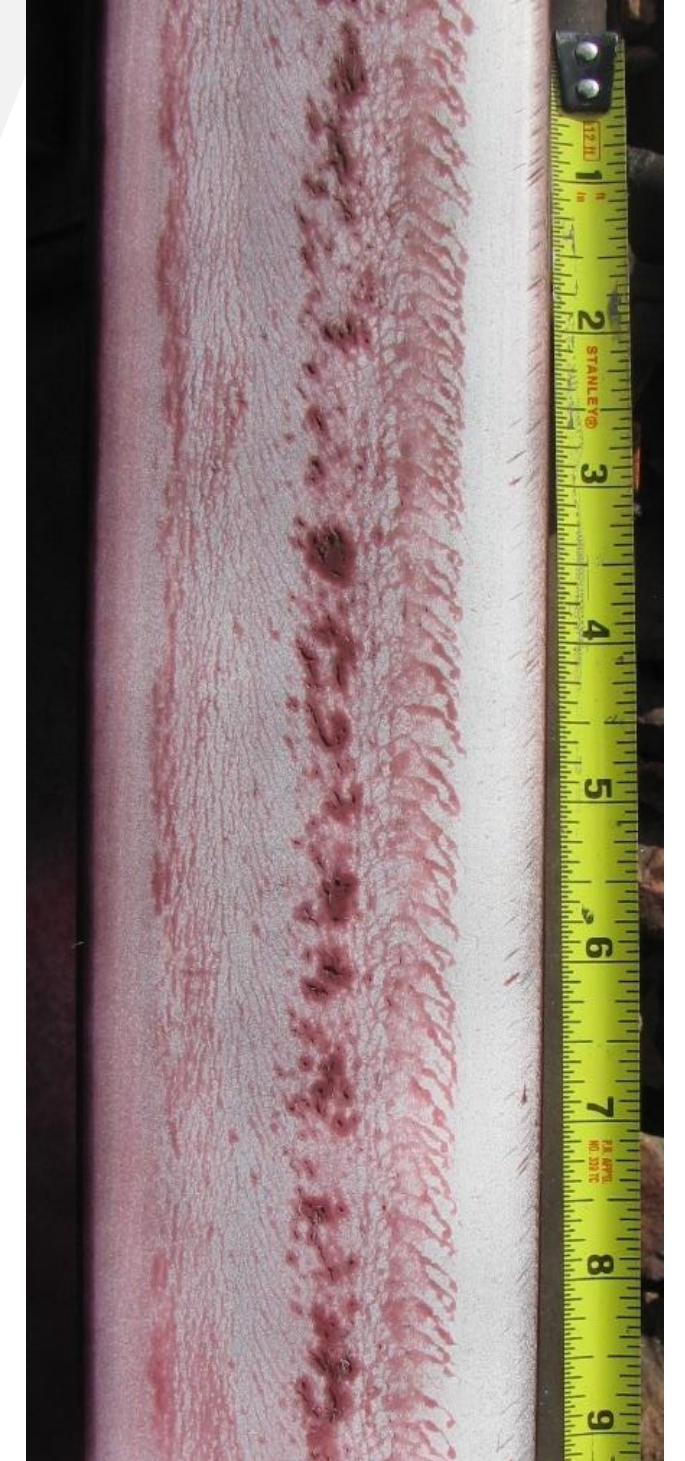


Concept of Magic Wear Rate by
Joe Kalousek and Eric Magel, 1997



Rail: Periodic Cracks

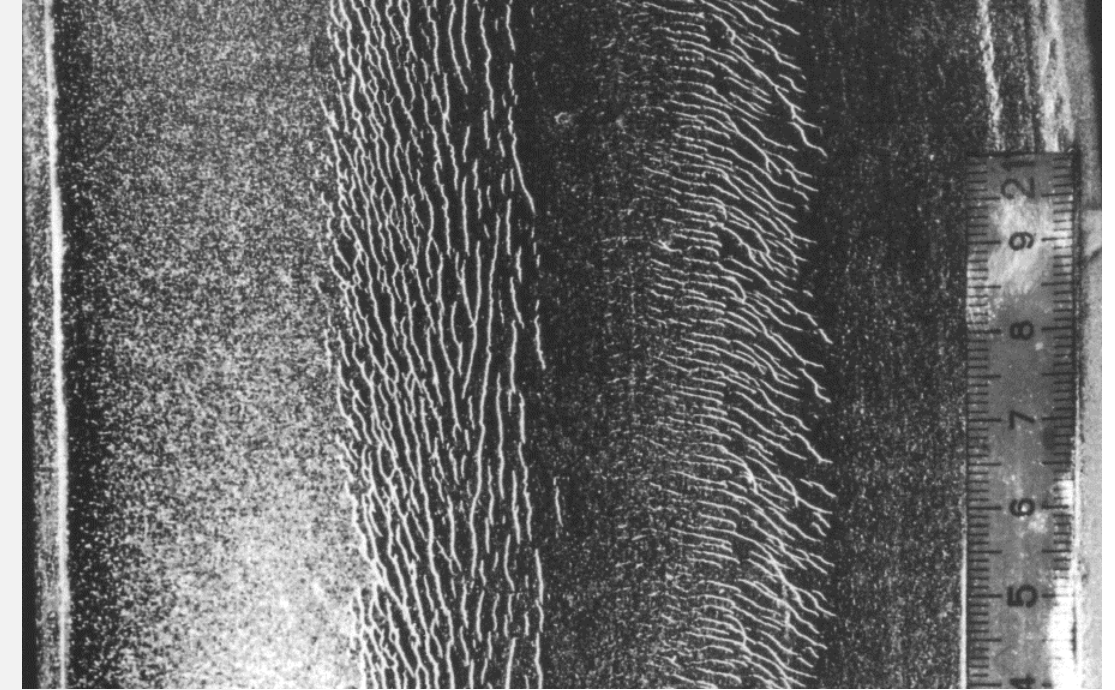
- Head Checks: periodic cracks at the gauge corner (gauge corner cracking)
- Heavy Haul: periodic cracks and crack networks also on the running surface
- Can cause detail fracture if not treated





Wheel: Periodic Cracks

- Rolling Contact Fatigue Cracks
- At various positions along the wheel profile
 - Flange side to field side



Source: E. Magel and J. Kalousek, 2004



Source: E. Magel and J. Kalousek, 2004



Source: Dedmon, CM2009



Source: E. Magel and J. Kalousek, 2004



Rail: Flaking and Spalling

- Head Checks can combine causing material to break out of the rail surface.
- Head Checks – Flaking – Spalling





Rail: Shelling

- Originates underneath the rail surface
- Delamination of rail material – crack will surface at gauge corner and cause break-outs
- High loading conditions favor formation – overstressing of subsurface area





Wheel: Shelling

- Different definition than rail
- Propagating cracks link beneath surface causing a large piece of material to spall off
- RCF cracks
- Thermo-mechanical (RCF) cracks
 - Material softening due to repeated heating



Source: E. Magel and J. Kalousek, 2004



Source: Eckberg and Kabo, 2005



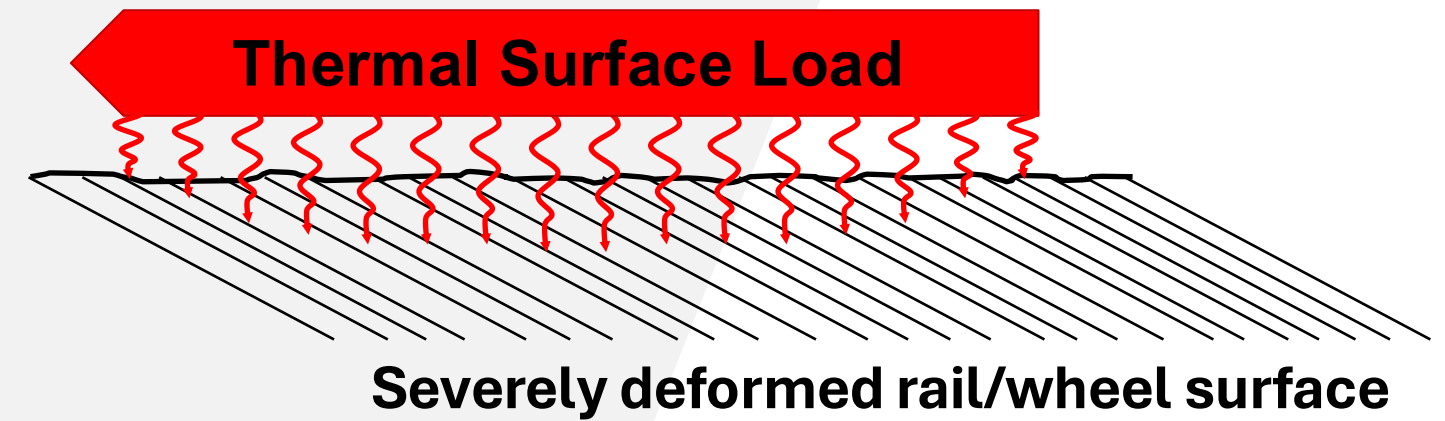
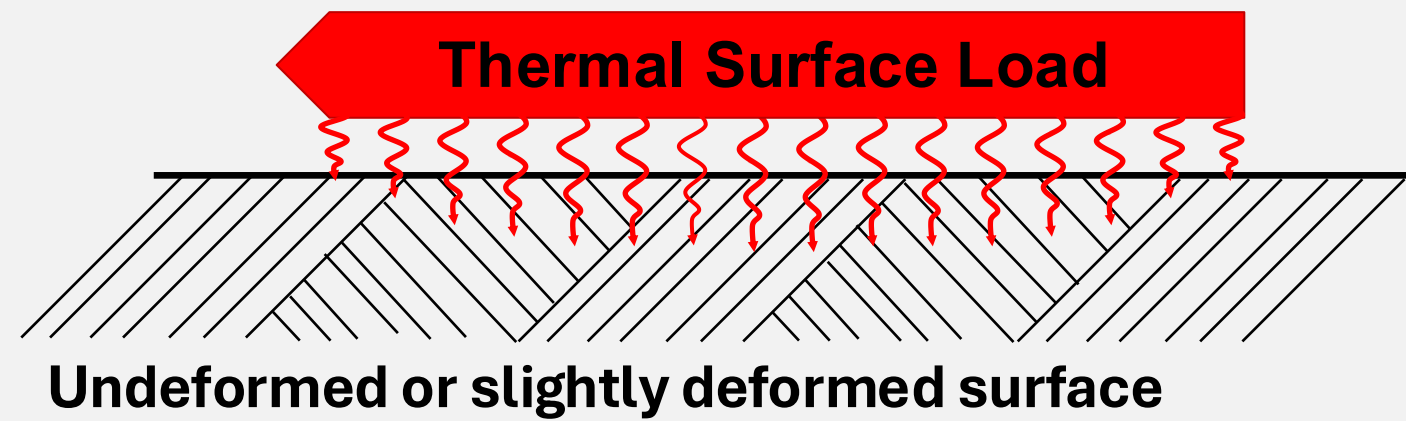
Rail: Squats

- Widening of running band / dip
- Typical kidney shaped
- Surface and subsurface crack(s)
- Singular or massed occurrence
- Characteristics
 - Heavily sheared rail surface
 - Crack initiation and growth by ratcheting (RCF)
 - Slow growth (within 100 MGT)
 - Can result in rail break





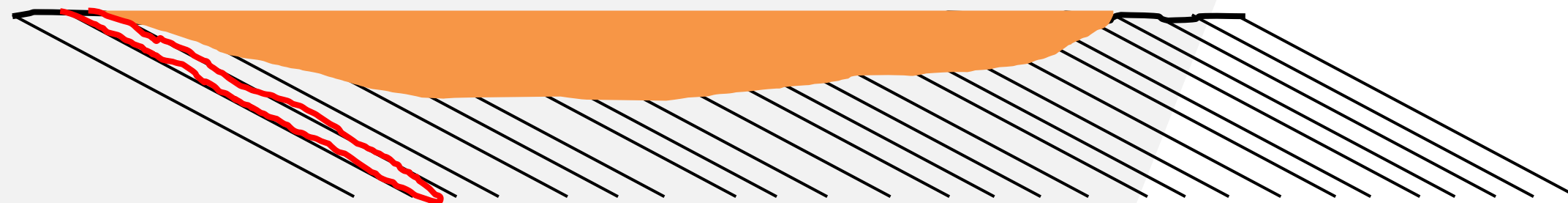
Material Response: Thermal Transformation



Quenching by rail/wheel mass (heatsink)



Material Transformation: White/Brown Etching Layer



Cracks might develop at interface and/or within layer

Schematic drawings

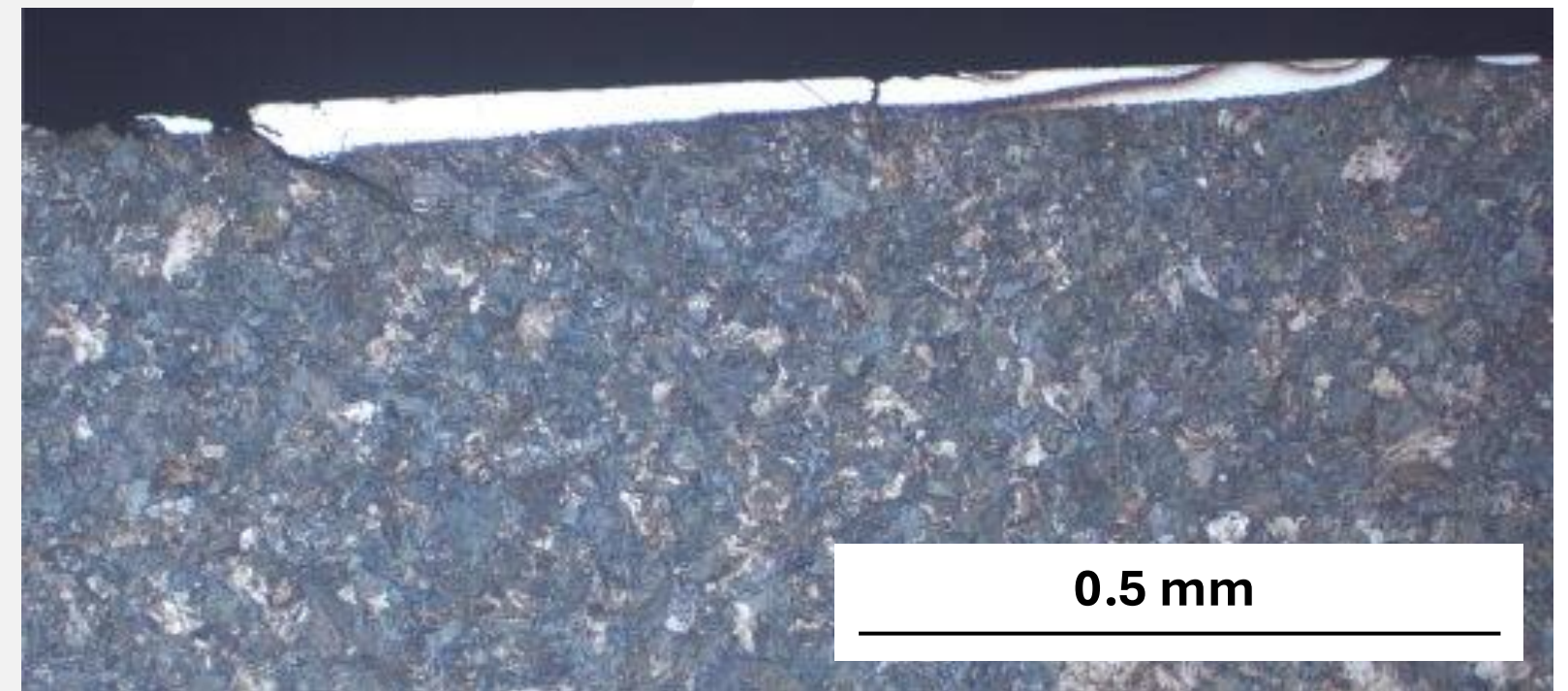
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DALLAS, TX



Thermal Transformation

- Thermal Transformation of rail or wheel surface
- Source: traction systems, rail maintenance, breaking
- Usually wears away
- White color in light microscope
 - Etching of surface to create contrast
 - 3% Nital Acid
 - Does not etch the Martensite / hard layers – appear white in LIM





Rail: Squat Type Defects / Studs

- Superficial similarity to Squats
- Mostly epidemic appearance
- Extended spalling of rail surface possible
- Characteristics:
 - Almost no plastic deformation
 - Associated with “white etching layers” (martensitic layers)
 - Formation within 10MGT or less



Photo: Rene Heyder, DB



Studs – A Complex Problem

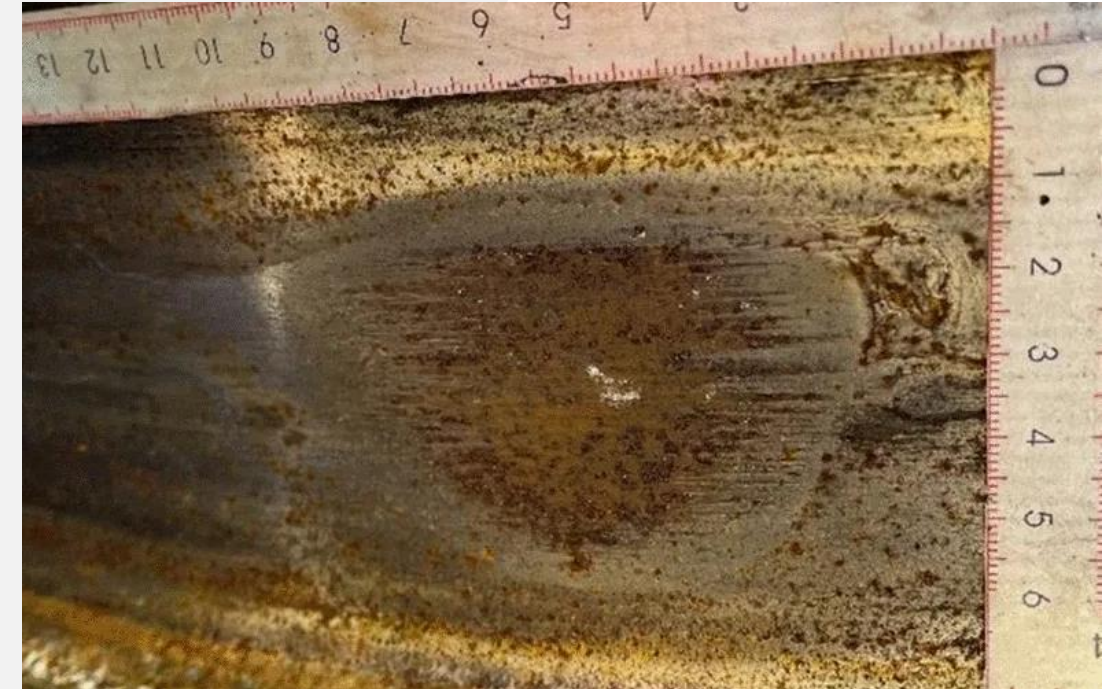
- Multiple contributing factors
 - Wear behaviour, R/W profiles, traction/friction conditions, system stiffness, rail maintenance activities
- Ongoing research and scientific debate



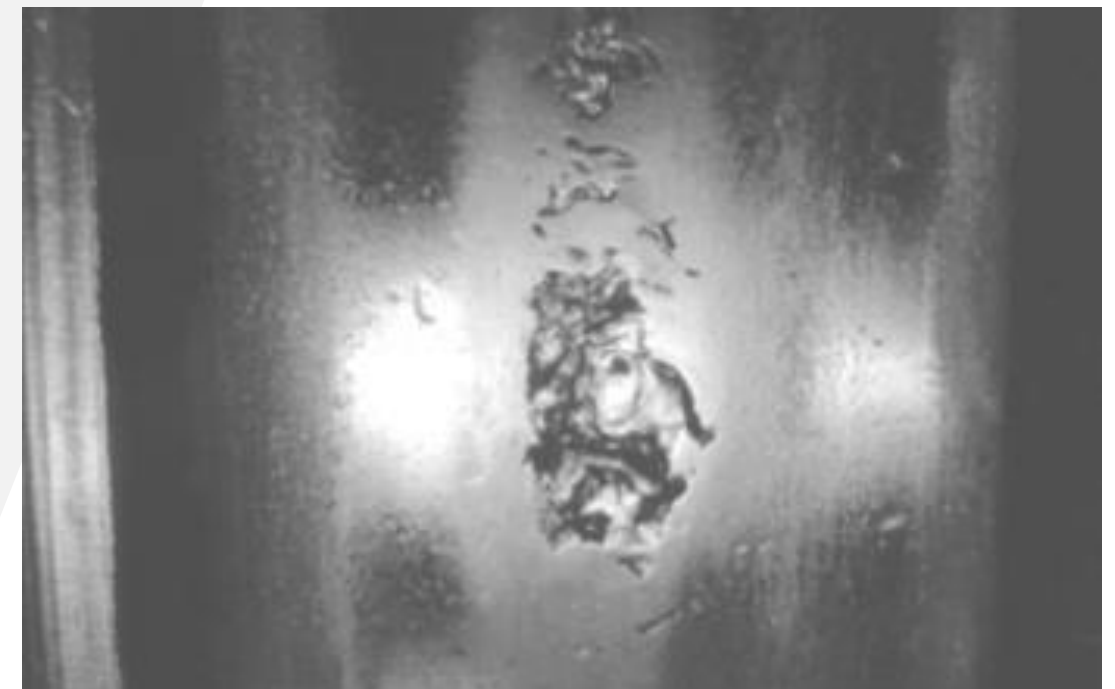


Wheel: Spalling

- Different definition than rail
- Martensite patch induced by wheel sliding (skid flat) – thermal damage
- Crack initiation and material to break out of surface – Spalling
- Boundary between Shelling (RCF) and Spalling (thermal damage) blurry



Source: Maglio et al, 2024



Source: Magel and Kalousek, 1996



Rail: Wheel Burn

- Occurs in pairs (both rails)
- Continuous slipping of locomotive wheel set(s).
- High temperature input to rail surface.
- Wear, plastic deformation, material transformation, cracks, break outs





Wheel: Wheel Flat

- Wheel sliding along the rail
 - Counterpart to Wheel Burn
- Wear, plastic deformation, heat induces material transformation (martensite), cracks, spalling



Source: Wikipedia, Bobo11,
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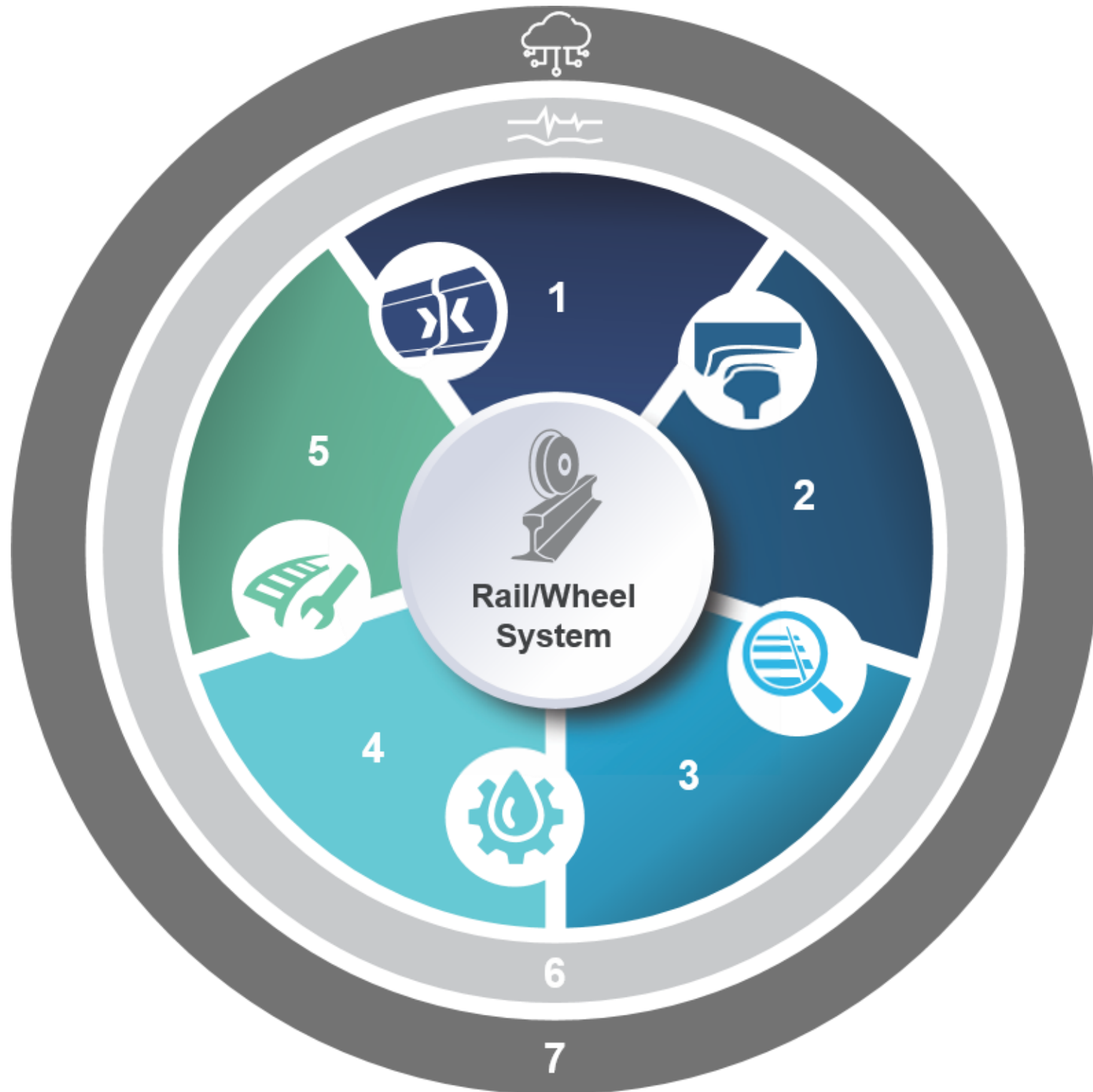
Controlling Rail and Wheel Damage

How to manage asset life





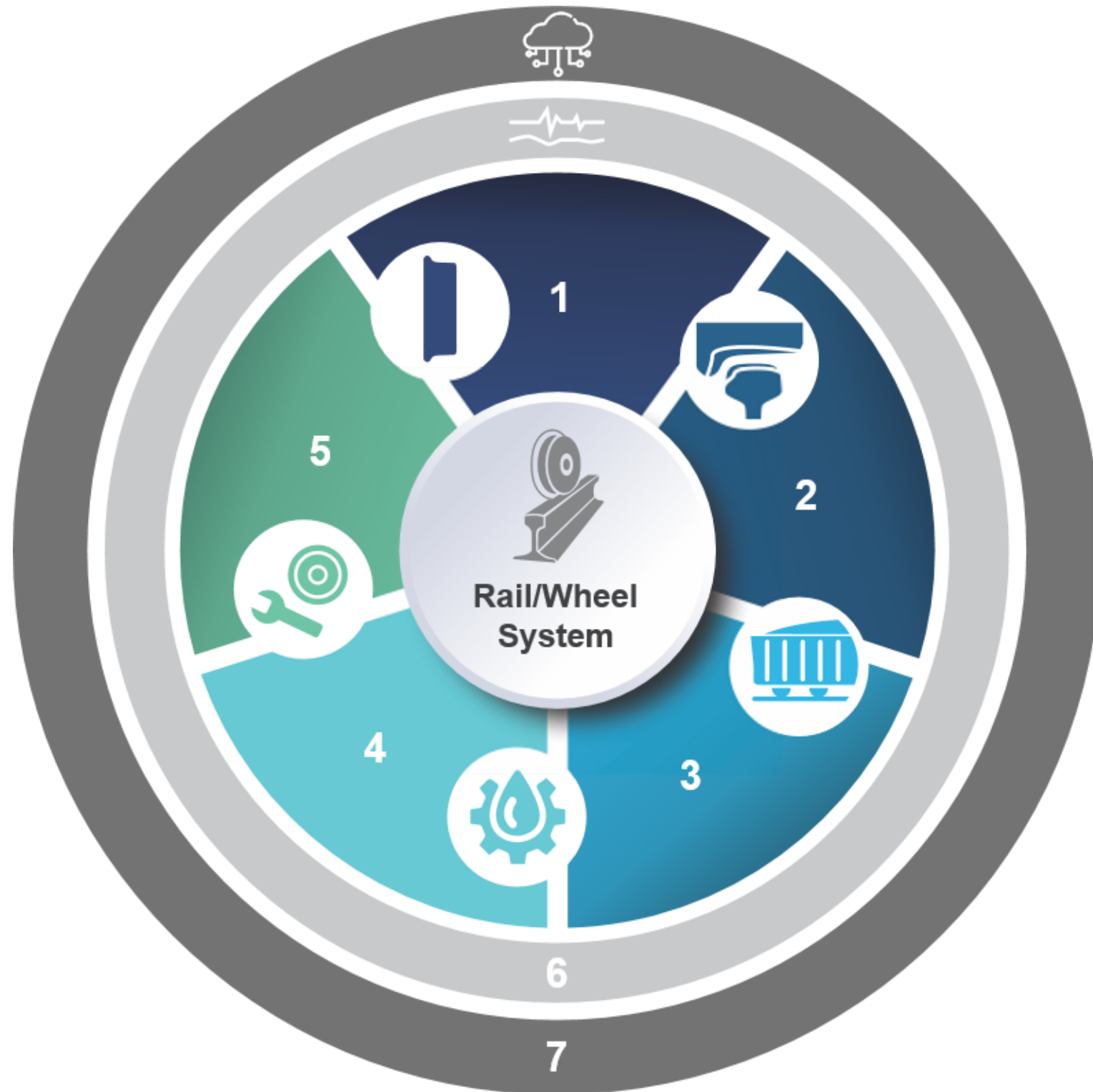
Mitigating Rail Damage



- 1 Material and Joining
- 2 Profiles
- 3 Track Quality
- 4 Friction Management
- 5 Rail Treatment / Rail Maintenance
- 6 Measurement Technology
- 7 Data Handling Technology



Mitigating Wheel Damage

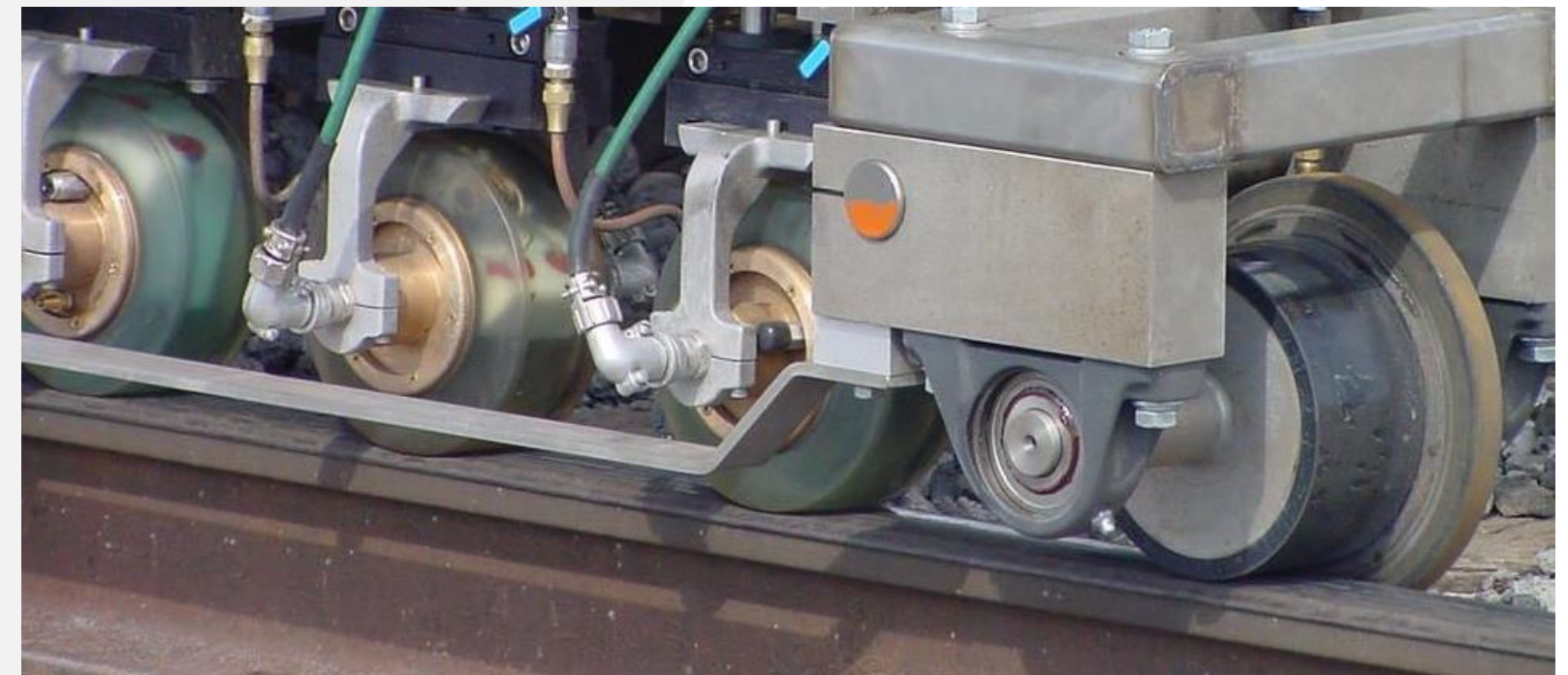
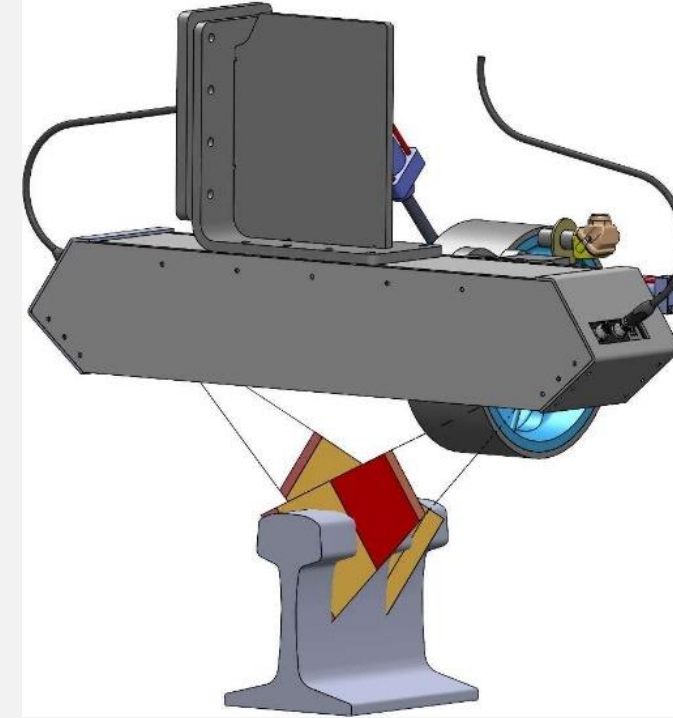


- 1 Wheel Materials
- 2 Wheel/Rail Profiles
- 3 Vehicle Condition
- 4 Friction Management
- 5 Wheel Maintenance
- 6 Measurement Technology
- 7 Data Handling Technology



Measurement Technology

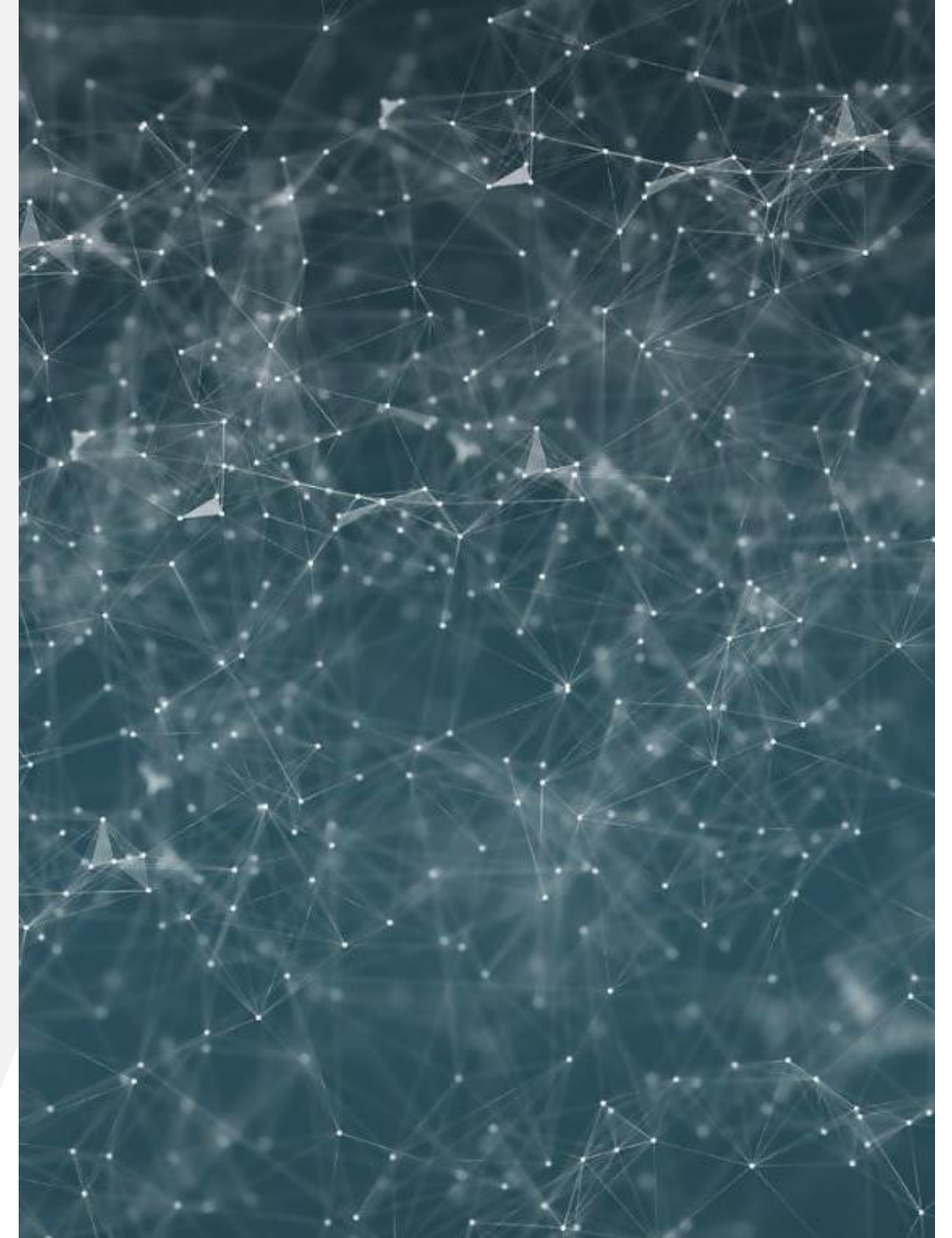
- Transversal Profile
- Longitudinal Profile / Roundness
 - Periodic and singular defects
- Visual Inspection
- Vision Technology
 - Surface features
- Electromagnetic Technology
 - Near surface damage
- Ultrasound Testing
 - Internal damage
- Hand tools, vehicle integrated, wayside systems





Data Handling & Management

- Requirements
 - Import from various data sources
 - Data integrity, integration and analysis
 - Data synchronisation between machines and measurement devices
 - Cyber-Security (Data, machines, measurement devices, back office)
- Visualisation and Reports
- Work order creation and transfer to maintenance machine





System Damage Mitigation – I

- Advanced Materials
 - Optimised material structure
 - Reduced wear and damage (both)
- Rail Welding
 - Prevention of premature damage on welds
- Optimised Profiles
 - Reduced contact stresses
 - Improved steering (curves) and stability (tangent)





System Damage Mitigation - II

- Track and vehicle condition
 - Dynamic system interaction
 - Mutual beneficial or detrimental effect
- Friction Management
 - Optimised friction between wheel and rail (TOR / GF)
 - Improved steering, reduced contact stresses / damage



Source: LB Foster



System Damage Mitigation - III

- Rail Maintenance
 - Grinding and Milling
- Wheel Maintenance
 - Wheel Turning
- **The only measure that can remove damage and not delay the formation!**



Source: Danobat



Rail Steel Grades

AREMA Chapter 4 Rail (2021)

EN 13674-1:2011

Carbon Standard Strength

Low Alloy Standard Strength

Carbon Intermediate Strength (HH)

Low Alloy Intermediate Strength

High Strength (HH)

Low Alloy High Strength (HH)

Steel grades according to AREMA Chapter 4 Rail (2021) and EN 13674-1:2011

	Chemical composition (%)						Mechanical data		
grade	C	Si	Mn	P _{max}	S	Cr	R _m [Ksi] min	Elong. [%] min	Hardness [HB]
R260	0.62-0.80	0.15-0.58	0.70-1.20	0.025	0.08-0.025	≤0.15	127	11	260-300
SS	0.74-0.86	0.10-0.60	0.75-1.25	0.020	0.020	0.30	142.5	10	310
LA	0.72-0.82	0.10-0.50	0.80-1.10	0.020	0.020	0.25-0.40	142.5	10	310
IS	0.74-0.86	0.10-0.60	0.75-1.25	0.020	0.020	0.30	155	10	350
IH	0.72-0.82	0.10-1.00	0.70-1.25	0.020	0.020	0.40-0.70	147	8	325
R350HT	0.72-0.80	0.15-0.58	0.70-1.20	0.020	0.025	≤0.15	170	10	350-390
R350LHT	0.72-0.80	0.15-0.58	0.70-1.20	0.020	0.025	≤0.30	170	10	350-390
HH	0.74-0.86	0.10-0.60	0.75-1.25	0.020	0.020	0.30	171	10	370
LH	0.72-0.82	0.10-1.00	0.70-1.25	0.020	0.020	0.40-0.70	171	10	370
R370CrHT	0.70-0.82	0.40-1.00	0.70-1.10	0.020	0.020	0.40-0.60	185	10	370-410
R400HT	0.90-1.05	0.20-0.40	1.20-1.30	0.020	0.020	≤0.30	185	10	400-440

Several Hypereutectoid grades and experimental Bainitic grades

Wheel Steel Grades

AAR M-107/M-208 (2020)

EN 13262:2021

No tension testing required

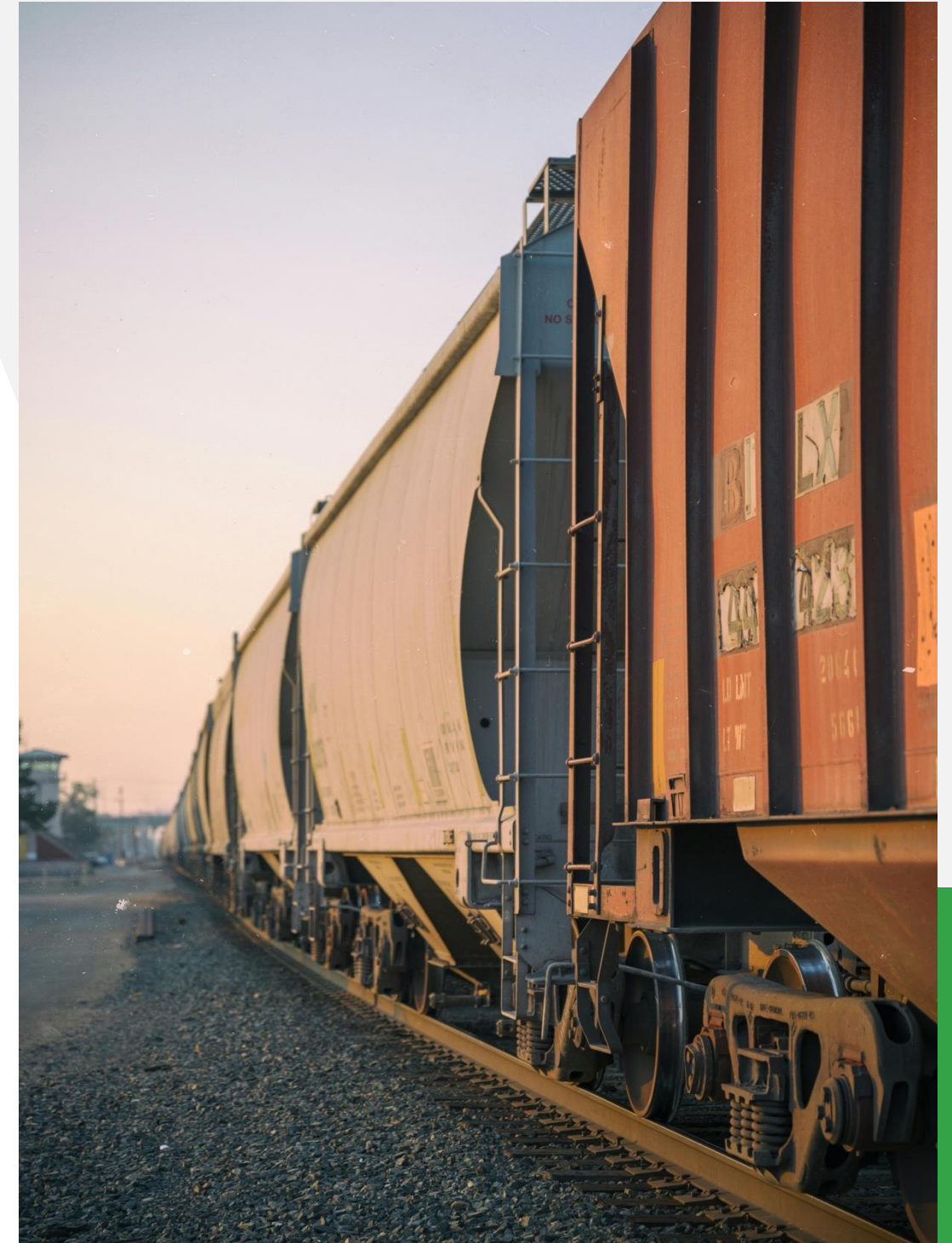
Microalloying Additions: Nb, V, Mo, Ni

Wheel Steel grades according to AAR M-107/M-208 and EN 13262:2021									
	Chemical composition (%)						Mechanical data		
grade	C	Si	Mn	P _{max}	S	Cr	R _m [MPa] min	Elong. [%] min	Hardness [HB]
ER6	0.48	0.40	0.75	0.02	0.015	0.30	780-900	15	225
ER7	0.52	0.40	0.80	0.02	0.015	0.30	820-940	14	235
ER8	0.56	0.40	0.80	0.02	0.015	0.30	860-980	13	245
ERS8	0.57	1.10	1.10	0.02	0.015	0.30	900-1020	13	250
ER9	0.60	0.40	0.80	0.02	0.015	0.30	900-1050	12	255
Class L	0.47	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	-	-	197-277
Class A	0.47-0.57	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	-	-	255-321
Class B	0.57-0.67	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	-	-	302-341
Class C	0.67-0.77	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	>1100	>14%	321-363
Class D	0.67-0.77	0.15-1.00	0.60-0.90	0.03	0.005-0.04	0.25	>1100	>14%	341-415



Summary - I

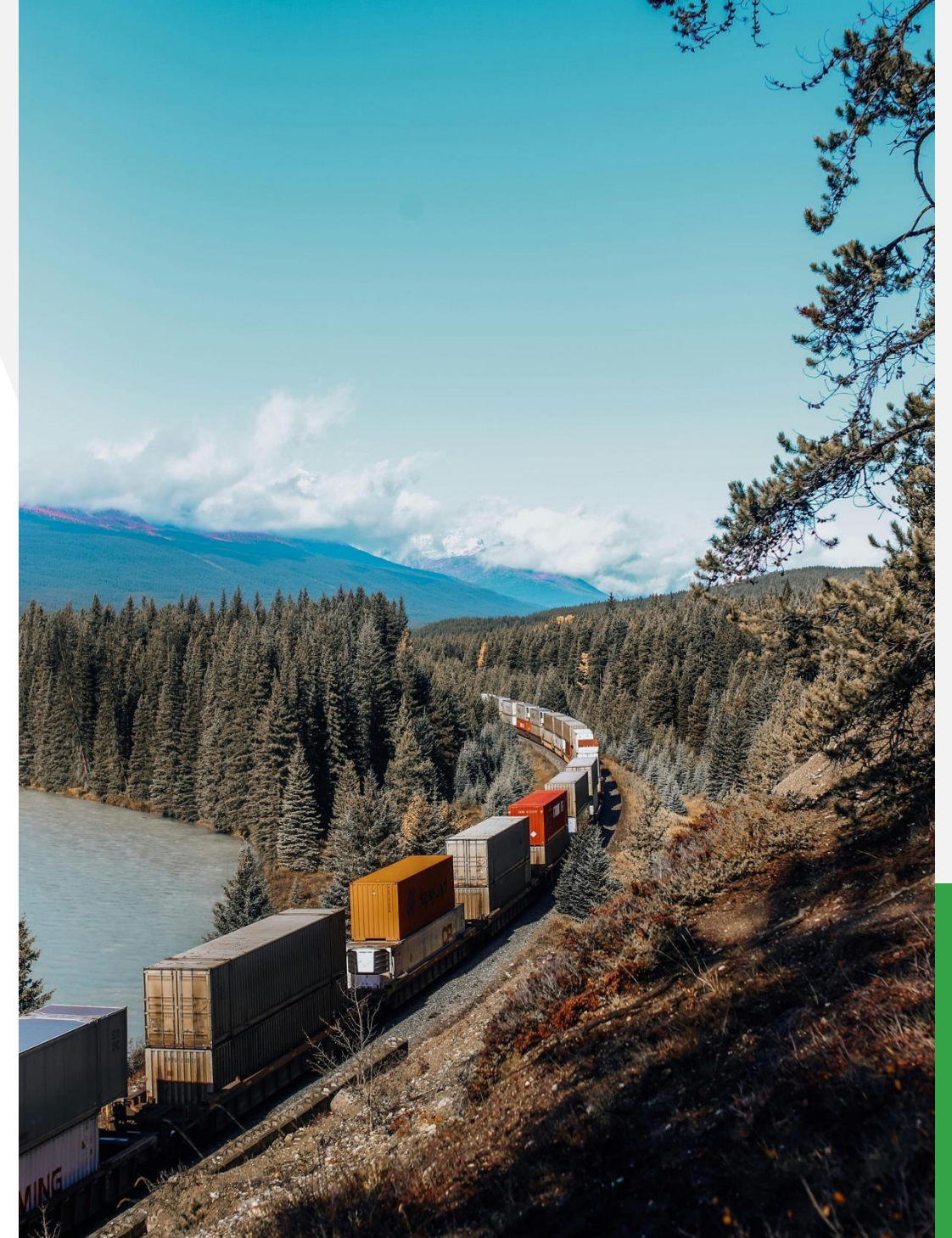
- Rail and wheel material
 - Typical rail/wheel steel: pearlitic steel
 - Lamella material structure
- Rail / wheel damage types
 - Plastic deformation, wear, cracks, thermal damage





Summary - II

- Controlling damage
 - Measurement Technology and Data handling
 - Material selection, wheel and rail profiles, track and vehicle condition, friction management, wheel and rail maintenance





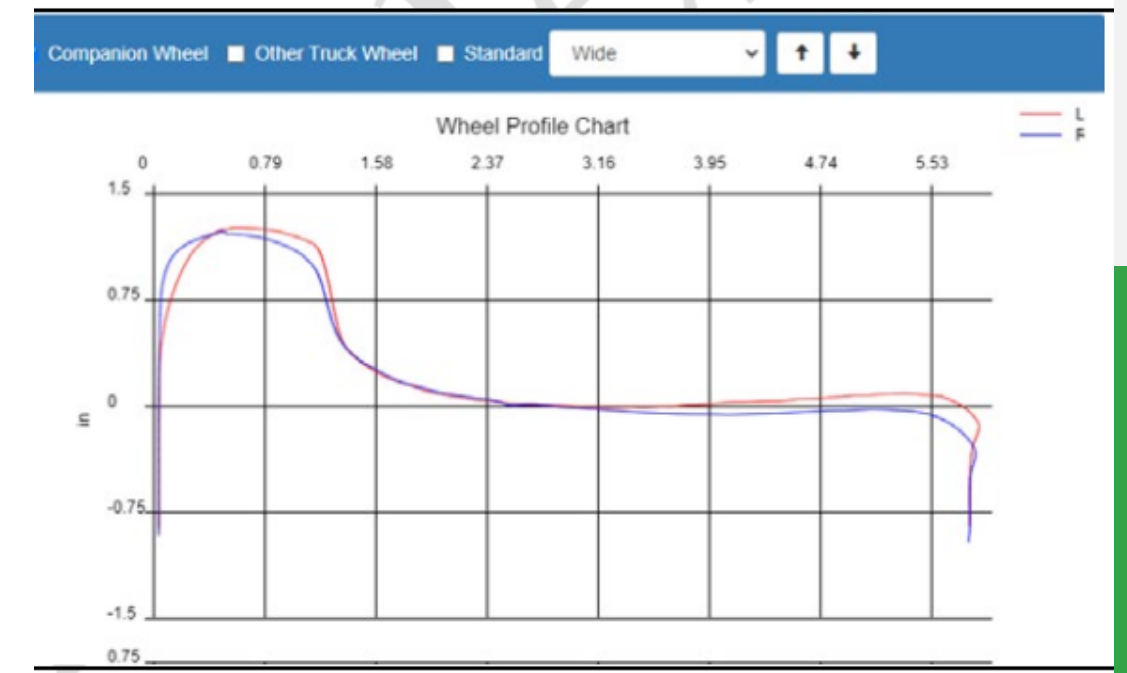
Thank you for your attention!

Plasser American



Back to the Derailment

- Rail Profile: AREA 141
- Rail grade: assumed Standard
- Wheel Profile: standard
- Wheel grade: no info
- Wheel wear:
 - Max. 2.5 mm hollow (limit 4mm)
- No other information





Rail Surface Condition





Wheel Surface Condition

